

Lab 3: Using EF Core CLI to Create and Apply Migrations

Scenario:

The retail store's database needs to be created based on the models you've defined. You'll use EF Core CLI to generate and apply migrations.

Objective:

Learn how to use EF Core CLI to manage database schema changes.

Source code

Migrations/20260706123045_InitialCreate.cs

```
using Microsoft.EntityFrameworkCore.Migrations;
namespace RetailInventory.Migrations
{
    public partial class InitialCreate : Migration
    {
        protected override void Up(MigrationBuilder migrationBuilder)
        {
            migrationBuilder.CreateTable(
                name: "Categories",
                columns: table => new
                {
                    Id = table.Column<int>(type: "int", nullable: false)
                        .Annotation("SqlServer:Identity", "1, 1"),
                    Name = table.Column<string>(type: "nvarchar(max)", nullable: true)
                },
                constraints: table =>
                {
                    table.PrimaryKey("PK_Categories", x => x.Id);
                });
            migrationBuilder.CreateTable(
                name: "Products",
```

```

columns: table => new
{
    Id = table.Column<int>(type: "int", nullable: false)
        .Annotation("SqlServer:Identity", "1, 1"),
    Name = table.Column<string>(type: "nvarchar(max)", nullable: true),
    Price = table.Column<decimal>(type: "decimal(18,2)", nullable: false),
    CategoryId = table.Column<int>(type: "int", nullable: false)
},
constraints: table =>
{
    table.PrimaryKey("PK_Products", x => x.Id);
    table.ForeignKey(
        name: "FK_Products_Categories_CategoryId",
        column: x => x.CategoryId,
        principalTable: "Categories",
        principalColumn: "Id",
        onDelete: ReferentialAction.Cascade);
});
migrationBuilder.CreateIndex(
    name: "IX_Products_CategoryId",
    table: "Products",
    column: "CategoryId");
}
protected override void Down(MigrationBuilder migrationBuilder)
{
    migrationBuilder.DropTable(name: "Products");
    migrationBuilder.DropTable(name: "Categories");
}
}
}

```

OUTPUT

```
Build started...  
Build succeeded.  
Applying migration '20260706123045_InitialCreate'.  
Done.
```

MigrationId	ProductVersion
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20260706123045_InitialCreate	8.0.10